SHRUTHI GOWDA

PhD Researcher | (Prev) AI Research Engineer

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RESEARCH PROFILE

PhD researcher focused on continual learning, representation learning, multimodal learning, and post-training adaptation of pretrained models. Research interests include robust generalization, efficient AI, and brain-inspired learning. Previously an AI research engineer developing computer vision and embedded AI systems for real-world applications.

RESEARCH INTERESTS

Continual & Lifelong Learning · Representation Learning · Multimodal Learning · Foundation Model Adaptation & Post-Training · Robustness & Generalization · Efficient AI

EDUCATION

PhD, Computer Science | Eindhoven University of Technology, Netherlands · *Jun 2024–present*

MSc, Electrical and Computer Engineering | UPC, Spain & UCLouvain, Belgium · *2010–2012*

BE, Electronics and Communication Engineering | Visvesvaraya Technological University, India · *2006–2010*

RESEARCH & PROFESSIONAL EXPERIENCE

AI / Computer Vision Research Engineer | NavInfo Europe, Netherlands · *2019–2024*

- Developed autonomous-driving perception and scene-understanding systems, privacy-preserving anonymization, and HD-map change detection.
- Led deep-learning and automotive use-case contributions in the SAFEXPLAIN European research project.

Senior Software Engineer, Vision R&D | National Instruments, India · *2013–2019*

- Developed and shipped machine-vision inspection algorithms; pioneered deep-learning-based defect detection and inference integration.
- Optimized computer vision and inference pipelines for CPU and embedded deployment.

Research Assistant | UCLouvain, Belgium · *2012–2013*

- Researched multi-view geometry, camera calibration, 3D vision, and virtual-viewpoint reconstruction.

SELECTED PUBLICATIONS

1. **Conceptual Plasticity for Online Domain-Incremental Vision Learning**
   Shruthi Gowda, Ashhar Aziz, Elahe Arani, Bahram Zonooz · *WACV 2027*
2. **TORINO: Token Reduction via Interpretable Concept Overlap in Vision-Language Models**
   Riccardo Renzulli, Gabriele Spadaro, Shruthi Gowda, Mazouz Alaa Eddine, Van-Tam Nguyen · *WACV 2027*
3. **Language Guidance for Supervised Vision Training: An Empirical Study of Generalization**
   Shruthi Gowda, Bahram Zonooz, Elahe Arani · *TMLR 2026*
4. **Conserve-Update-Revise to Cure Generalization and Robustness Trade-off in Adversarial Training**
   Shruthi Gowda, Elahe Arani, Bahram Zonooz · *ICLR 2024*
5. **Can We Break Free from Strong Data Augmentations in Self-Supervised Learning?**
   Shruthi Gowda, Elahe Arani, Bahram Zonooz · *CoLLAs 2024*
6. **Dual Cognitive Architecture: Incorporating Biases and Multi-Memory Systems for Lifelong Learning**
   Shruthi Gowda, Elahe Arani, Bahram Zonooz · *TMLR 2023*

7. **InBiaseD: Inductive Bias Distillation to Improve Generalization and Robustness through Shape-awareness**
   Shruthi Gowda, Elahe Arani, Bahram Zonooz · *CoLLAs 2023*
8. **A Comprehensive Study of Real-Time Object Detection Networks Across Multiple Domains: A Survey**
   Shruthi Gowda et al. · *TMLR 2022 · Survey Certification*

SELECTED RESEARCH & ENGINEERING PROJECTS

ENFIELD | Horizon Europe · Adaptive, Green & Trustworthy AI ·

- Research on efficient continual learning, robustness, and generalization in a European multi-partner AI initiative.

SAFEXPLAIN | Horizon Europe · Safety & Explainability in Embedded AI ·

- Led deep-learning and automotive use-case work on explainable components and safety-aware design for autonomous driving.

Autonomous Driving Perception & HD Mapping | NavInfo Europe ·

- Developed multi-camera scene understanding, anonymization of faces and license plates, and infrastructure-change detection from geo-tagged video.

Industrial Machine Vision & Deep Learning | National Instruments ·

- Developed pattern matching, OCR, barcode reading, defect inspection, and optimized deep-learning inference for industrial systems.

PATENTS & PATENT APPLICATIONS

- European application 22159288.4 — AI-based change detection from geo-tagged video for HD-map updates.
- US application 17/894,870 — Shape awareness in self-supervised learning.
- Dutch filing 020065 NL-PD — Leveraging shape information in few-shot learning.
- US filing 019922 US-PD-VB/kk — Inductive bias in deep neural networks.
- US application 17/581,759 — Deep-learning-based multi-sensor detection system.
- Dutch filing 020050 NL-PD1 — Cognitive continual learning system.

TECHNICAL EXPERTISE

Deep Learning: PyTorch, TensorFlow, ONNX, CLIP, Vision Transformers | Programming: Python, C/C++, MATLAB, LabVIEW | Vision & Deployment: OpenCV, OpenVINO, Intel IPP, DAAL, TBB, TensorRT

SELECTED AWARDS

NI Engineering Excellence Award (2016) · NI Rookie of the Year (2013) · Pratibha Puraskar (2006)